



University of Central Punjab

(Incorporated by Ordinance No. XXIV of 2002 promulgated by Government of the Punjab)
FACULTY OF INFORMATION TECHNOLOGY

Data Structure & Algorithm -LAB

Mid Term Spring 2026

Paper Duration: 90 minutes

Total Marks: (10+10 = 20)

INSTRUCTIONS:

- Midterm is self-explanatory, no query will be entertained.
- Submit only **.cpp files**. (In case of multiple files, compressed into zip folder)
- 50% deduction in case of run time error (**segmentation fault**).
- 70% deduction in case of **Logical error**.

Question No: 1

Marks -10

Online Food Delivery Management System

You are hired to develop a system for an online food delivery service. The system must manage incoming orders and process them efficiently. Customers place orders continuously, and each order must be handled in **First-Come, First-Served (FCFS)** order. However, the system also needs flexibility to manage and update order records dynamically

System Requirements:

Each order contains:

- Order ID (int)
- Customer Name (char array)
- Food Item (string)
- Priority Level (1 = Urgent, 2 = Normal)

Implementation Details:

- Use a **Singly Linked List** to store all order records dynamically.
- Use a **Queue** to process orders in FCFS order.

Required Operations:

Linked List Operations:

1. Add a new order at the **end**
2. Delete an order by **Order ID**
3. Display all orders
4. Search order by ID



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Queue Operations :

5. Enqueue order (add to processing queue)
6. Dequeue order (order delivered)
7. Display current queue
8. Peek front order

Advanced Task:

9. Prioritize urgent orders:
 - o Move all **priority = 1** orders to the front of the queue
 - o Maintain relative order among same priority

Question No: 2

Marks -20

Book Pile System

You are managing a pile of books where only the **top book** can be added or removed. The system follows **Last-In, First-Out (LIFO)**.

Each book contains:

- Book ID
- Book Name

1. **Tasks:**
2. **displayBooks ()** – Display all boosearchBook (int id)
 - a. Search a book by ID in the stack
 - b. Return its position from top
3. **pushBook ()** – Add a book
4. **popBook()** – Remove top book
5. **peekBook ()** – Show top book
6. **countBooks ()** Return total number of books
7. **reverseStack ()** Reverse the stack using another stack
8. **findTopMostEvenID ()** Find the nearest book (from top) having even ID